

MERN Stack Syllabus

Course Duration

- **Duration:** 6 Months
 - **Level:** Beginner to Advanced
 - **Projects:** Mini Projects + 3 Major Projects
 - **Technologies:** HTML, CSS, JavaScript, React, Redux, Node.js, Express.js, MongoDB, Git, GitHub, Deployment
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Month 1 — Web Development Fundamentals

Module 1: Introduction to Web Development

- Internet Basics
- Client & Server
- Frontend vs Backend
- MERN Stack Overview
- VS Code Setup
- Browser Developer Tools

Module 2: HTML5

- HTML Structure
- Forms & Input Types
- Tables & Lists
- Semantic Tags

Module 3: CSS3

- Selectors
- Box Model
- Flexbox
- Grid
- Responsive Design
- Animations & Transitions

Module 4: Tailwind CSS

- Utility Classes
- Layout Building
- Responsive Utilities
- Component Styling

Mini Project 1

Responsive Portfolio Website

Features:

- About Section
- Skills Section
- Projects Section
- Contact Form
- Responsive Layout

Month 2 — JavaScript Development

Module 5: JavaScript Basics

- Variables
- Data Types
- Operators
- Conditions
- Loops
- Functions

Module 6: Advanced JavaScript

- DOM Manipulation
- Events
- ES6 Features
- Arrow Functions
- Destructuring
- Callbacks
- Promises
- Async/Await
- Fetch API
- Local Storage

Module 7: Git & GitHub

- Git Installation
- Git Commands
- Repository Creation
- Push/Pull

- Branching
- Pull Requests

Mini Project 2

Weather Application

Features:

- Weather API
- Search Functionality
- Dynamic Data
- Responsive UI

Month 3 — React.js Development

Module 8: React Fundamentals

- Introduction to React
- JSX
- Components
- Props
- State
- Event Handling

Module 9: React Hooks

- useState
- useEffect
- useRef
- useContext

Module 10: React Routing

- React Router DOM
- Dynamic Routes
- Navigation

Module 11: Form Handling & API Integration

- Controlled Components
- Form Validation
- Axios
- Fetch API
- Loading States

Module 12: Redux Toolkit

- Redux Concepts
- Store
- Reducers
- Actions
- Slices
- Global State Management

Mini Project 3

Movie Search Application

Features:

- API Fetching
- Search System
- Routing

- Favorites Feature
- Responsive Cards

Month 4 — Backend Development

Module 13: Node.js

- Node Basics
- npm
- Modules
- File System
- Event Loop

Module 14: Express.js

- Express Server
- Routing
- Middleware
- REST API Development

Module 15: MongoDB

- Database Basics
- Collections & Documents
- CRUD Operations
- MongoDB Atlas

Module 16: Authentication & Security

- JWT Authentication
- Password Hashing

- Protected Routes
- Role-Based Access

Module 17: Frontend & Backend Integration

- Connecting React with Backend
- API Communication
- Environment Variables
- Error Handling

Mini Project 4

Task Management System

Features:

- User Authentication
- CRUD Operations
- Redux Toolkit
- JWT Security
- MongoDB Database

Month 5 — Full Stack MERN Projects

Module 18: Deployment & Hosting (Free Hosting)

Frontend Hosting

- Vercel
- Netlify

Backend Hosting

- Render
- Railway

Database Hosting

- MongoDB Atlas

Deployment Topics

- Environment Variables
- Build Process
- Production Deployment
- Custom Domains

Major Project 1

Topic : Skill Exchange Platform

Project Overview

A platform where users can exchange skills with others.

Example:

- One user teaches Graphic Design
- Another user teaches Programming
- Both can connect and learn from each other

Features

- User Authentication
- User Profiles
- Skill Listings
- Search Skills
- Request & Accept Skill Exchange
- Chat System
- Ratings & Reviews
- Admin Dashboard

Month 6 — Advanced Projects & Career Preparation

Major Project 2

Topic : Home Chef Food Marketplace

Project Overview

A marketplace where home chefs can sell homemade food products online.

Features

- Chef Registration
- Food Product Listings
- Cart System
- Order Management
- Search & Categories
- Customer Authentication
- Delivery Address Management
- Admin Panel

Major Project 3 — Custom Capstone Project

Topic : Student Innovation Project

Students can choose and build their own project idea.

Example Ideas

- Learning Management System
- AI-Based Web App
- Food Delivery Platform
- Social Media Platform
- Event Management System
- Smart Farming Platform